

CEP I.1 | Campus Network Link Design

One-page roadmap summary

Problem

Connect **six campus buildings** (40 m to 1.8 km apart) with links that meet a target speed and uptime. One link crosses an **EMI-heavy corridor**, one crosses **open ground with no conduit**, and the **fixed budget cannot cover fibre everywhere**. We must choose the media and line code for each link and prove it works.

Solution and approach

A **Python link-budget model** tests every cable option for every link (signal loss, noise, Shannon and Nyquist limits, uptime, cost), rejects failures with reasons, and picks the cheapest option that passes. A **sensitivity test** then cuts the budget 15% to find the first link to fail. A **Packet Tracer network** proves reachability and duplex settings for the final plan.



Team roles

Group	Owns
Python pair	Link-budget calculations, line-code choice, availability, cost and media selection, 15% sensitivity analysis, code checking
Packet Tracer pair	IP plan, topology, applying the media plan, reachability and duplex testing
Documentation member	Report, assumptions, figures and tables, written alongside every stage

Roadmap by milestone

Milestone	Work planned	Output
M0 Kickoff	Form group, agree roles, open contribution log	Allotment signed, log opened
C1 Framing	Fix the six links, media catalogue, budget and assumptions; set up shared workspace (Steps 1 to 4)	Problem-framing note, work plan
C2 Design	Build the calculation model and option table; plan the network on paper (Steps 5 to 11, 14)	Design document: model, alternatives, tool choice
C3 Build	Sensitivity analysis, code checks, Packet Tracer build and first validation (Steps 12, 13, 15 to 17)	Working .pkt, first results
C4 Final	Integrate, write report, review and rehearse viva (Steps 18 to 20)	Final report, .pkt, code, viva

Final deliverables

Per-link table (loss, noise, SNR, Shannon, Nyquist, line code, uptime, cost) • media plan with rejected alternatives • Packet Tracer file with ping and duplex evidence • 15% budget-cut result and sweep plot • code, CSV data and final report.

Key rule: Python decides what to build; Packet Tracer proves it works as a network. They share one file, *results/media_plan.csv*.

Full 14-page project guide: _____